

Boards API — Mobile Developer Guide

Everything needed to build the Boards feature: the boards list screen, the board detail screen, and tapping a tile to control a device.

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What changed in 1.2. `/dashboards/config` now returns `entity_meta` — a per-entity record with the real Home Assistant icon, display name, category, live state and whether the tile is tappable — so you no longer derive icons or names from the entity ID. Three query parameters trim the board to what HA itself treats as dashboard tiles, and **voice-gated entities are now withheld from boards**, so every tile is a plain button click with no 409 to design around (§4). The new `GET /voice-gated` answers “which entities return 409?” in one read-only call instead of firing commands to find out (§5). §4 also corrects the status code for an unknown `url_path`.

1. Basics

Base URL

```
https://voiceorchestrator.homeadapt.us/api/v1/voice-auth
```

Authentication

Every request carries the device API key as a Bearer token. Keys are issued per device (prefix `sk_ios_`). Never embed a key in the app binary; store it in Keychain/Keystore after provisioning.

```
Authorization: Bearer sk_ios_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

Identity parameters

All Boards calls identify the caller with two values the app already holds after login: `user_ref` (e.g. `scott_mobile`) and `home_id` (e.g. `scott_home`). Dashboards are home-level in Home Assistant, so every user of a home sees the same board list; `user_ref` is required for auth and audit parity.

Error envelope

Every non-2xx response has the same shape:

```
{ "error": "human readable message", "code": "MACHINE_CODE" }
```

2. Data model in one paragraph

A home has **dashboards** (we call them boards). Each board has **views** — the rooms/tabs. Each view has **entities** — the devices, identified by Home Assistant entity IDs like `light.desk_lamp`. The part before the dot is the **domain**. All of this is configured in the home's Home Assistant; the API reads it live. The app never stores room structure. Alongside the entity IDs, `/dashboards/config` returns `entity_meta`: everything needed to draw the tile, resolved server-side against HA.

3. Screen 1 — Boards list (overall view)

One call populates the whole screen. The default whole-home board (`url_path = null`) is always first.

Request

```
GET /dashboards?user_ref=scott_mobile&home_id=scott_home
Authorization: Bearer <API_KEY>
```

Response — 200

```
{
  "home_id": "scott_home",
  "count": 6,
  "items": [
    { "url_path": null,          "title": "Overview",          "icon": "mdi:view-dashboard",
      "mode": "storage", "show_in_sidebar": true, "require_admin": false, "is_default": true },
    { "url_path": "map",        "title": "Map",              "icon": "mdi:map",
      "mode": "storage", "show_in_sidebar": true, "require_admin": false, "is_default": false }
  ],
  { "url_path": "scott-s-view", "title": "Scott's Office", "icon": "mdi:account-plus",
    "mode": "storage", "show_in_sidebar": true, "require_admin": false, "is_default": false }
}
```

Field	Type	Meaning
<code>url_path</code>	string?	Board identifier. null means the default whole-home Overview. Pass this to the config endpoint.
<code>title</code>	string?	Display name. Fall back to “Overview” when null.
<code>icon</code>	string?	Material Design icon name (mdi:*). Use your own icon set if null.
<code>mode</code>	string	HA storage mode. Informational only.
<code>show_in_sidebar</code>	bool	Whether HA shows it in its own sidebar. You may use it to hide utility boards.
<code>require_admin</code>	bool	Admin-only board in HA. Hide from non-admin users.
<code>is_default</code>	bool	true only for the whole-home Overview (always the first item).

Rendering rules (match the prototype)

One row per item: house icon when `is_default` is true, person icon otherwise. Title on top, `url_path` (or “whole home”) as the subtitle. Tapping a row opens Screen 2 with that item's `url_path`.

Errors

Status / code	When	App behavior
400 VALIDATION	<code>user_ref</code> or <code>home_id</code> missing	Fix the request; not user-visible.
404 NOT_CONFIGURED	<code>home_id</code> unknown to the platform	Re-run provisioning / sign-in.
503 HOME_UNREACHABLE	The house's HA is offline or unreachable	Show the cached list with a “Home unreachable” banner (see §6).
502 HA_ERROR	HA answered with an error	Show cached list + banner; retry later.
503 NOT_CONFIGURED	Server has no dashboard client wired	Treat as outage.

4. Screen 2 — One board (views + tiles)

One call returns every view (room tab) of the board, the entity IDs each view shows, and a metadata record per entity. Omit `url_path` for the default Overview board.

Request

```
GET /dashboards/config?user_ref=scott_mobile&home_id=scott_home
    &include_categories=primary&include_hidden=false
Authorization: Bearer <API_KEY>
```

Query parameters

Parameter	Default	Meaning
url_path	null	Which board. Omit for the whole-home Overview.
include_categories	all	Comma list of primary , config , diagnostic . A tile UI wants primary . An unknown value is a 400, not a silent no-op.
include_hidden	true	Send false to drop entities the user hid in HA.
include_gated	false	Voice-gated entities are withheld by default so every tile is a plain button click. Send true only if you are building a screen that handles the voice flow.
include_config	false	Adds the raw Lovelace JSON. Large and frontend-specific; the prototype never needs it.

Response — 200

```
{
  "home_id": "scott_home",
  "url_path": null,
  "title": null,
  "strategy": "original-states",
  "view_count": 16,
  "views": [
    { "title": "Deck", "path": "deck", "icon": null, "entity_count": 1,
      "entities": ["switch.deck_light"] },
    { "title": "Den", "path": "den", "icon": null, "entity_count": 4,
      "entities": ["light.den_lamp", "media_player.den_tv", "..."] }
  ],
  "entities": ["switch.deck_light", "light.den_lamp", "..."],
  "entity_count": 108,
  "entity_meta_available": true,
  "gated_excluded": ["lock.yale_yrd226_tsdb"],
  "gate_check_available": true,
  "entity_meta": {
    "switch.deck_light": {
      "name": "Deck light",      "domain": "switch",
      "icon": "mdi:toggle-switch", "icon_source": "domain",
      "device_class": null,      "unit_of_measurement": null,
      "entity_category": null,    "hidden": false,
    }
  }
}
```

```

        "state": "off",                "controllable": true,
        "voice_gated": false,         "voice_auth_enrollment_id": null
    },
    "sensor.deck_light_signal_level": {
        "name": "Deck light Signal level", "domain": "sensor",
        "icon": "mdi:signal",            "icon_source": "translation",
        "device_class": null,            "unit_of_measurement": null,
        "entity_category": "diagnostic", "hidden": false,
        "state": "2",                    "controllable": false,
        "voice_gated": false,            "voice_auth_enrollment_id": null
    }
}
}
}

```

Field	Type	Meaning
title	string?	Board display name for the header.
strategy	string?	Set (e.g. "original-states") when the board is auto-generated by HA. The server already expands it into one view per area — render exactly like a normal board.
views[].title	string?	Tab label. Fall back to path.
views[].path	string?	Stable view identifier. Persist it to reopen the same tab.
views[].entity_count	int	Post-filter count. Show in the tab chip, e.g. "Den · 4".
views[].entities	string[]	HA entity IDs, in display order. One tile each.
entities	string[]	All unique entity IDs across views (deduplicated).
entity_meta	object	entity_id → tile record. Keys match entities exactly.
entity_meta_available	bool	false means HA served the board but not the metadata — entity_meta is empty and the filters were not applied. Fall back to the domain table below.
gated_excluded	string[]	Entities held back because they need voice auth. Usually empty. They are not tiles — reach them through Favorites.
gate_check_available	bool	false means the enrollment lookup failed, so gated entities could not be held back and a tap may still 409. Keep the handler for this case.

A board tile is always a button click

Voice-gated entities are withheld from a board by default, so every tile `/dashboards/config` returns can be tapped and will fire. There is no voice badge to render on this screen and no 409 to design around: tap → §5 → 200. Anything held back is named in `gated_excluded` so the screen can say "1 device needs voice authentication" if it wants to, and it stays reachable through Favorites, which runs the voice flow.

This does not weaken the gate — a withheld device still requires the spoken challenge everywhere it IS offered. On the reference home exactly one entity is affected, the front-door lock, which is why the Entry view shows one tile rather than two. Keep your 409 handler for the `gate_check_available: false` case and for someone enrolling a device between your fetch and the tap.

The entity_meta record

Field	Meaning
name	Display name, straight from HA. Use it verbatim — do not title-case the entity ID.

Field	Meaning
icon	Material Design icon name, always populated. Resolved against HA itself (see below).
icon_source	Which rung of the chain produced the icon: state, registry, integration, translation, device_class, unit, domain. Diagnostic only — do not branch on it.
device_class	HA device class when there is one (temperature, door, tv...). Useful for choosing a value format.
unit_of_measurement	Unit to append when showing state, e.g. “°F”.
entity_category	null (a primary dashboard entity), config , or diagnostic .
hidden	true when the user hid the entity in HA.
state	Live state at request time. Server caches HA state for 10 seconds.
controllable	true when tapping can fire the entity through \$5. false → render read-only.
voice_gated	Always false on a default board — gated entities are withheld, so this only ever reads true when you asked for <code>include_gated=true</code> . When it is true, a tap WILL return 409; route it to the VAPI session (\$5).
voice_auth_enrollment_id	The enrollment doing the gating, or null. Useful for logs and for “manage voice authentication”.

Why the icon cannot come from the entity ID

Home Assistant stopped putting icons on entities. Since HA 2024.2 an integration declares them in an icon-translation file keyed by an internal translation key, which the HA REST API does not expose at all. On a real home, 62% of the entities on the default board carry neither an icon nor a device class over REST — `sensor.deck_light_signal_level` is one of them, yet HA shows `mdi:signal` for it. The server now reads the entity registry and the icon translations over HA's WebSocket API and resolves each entity through the chain below, so a tile matches what the user sees in Home Assistant. Every entity gets an icon; there is no null case to handle.

#	Source	What it is
1	state	The entity's live icon attribute.
2	registry	An icon the user set by hand in HA.
3	integration	A static icon shipped by the integration.
4	translation	The <code>icons.json</code> entry — the modern path, and the largest single source.
5	device_class	Our map, e.g. <code>temperature</code> → <code>mdi:thermometer</code> .
6	unit	Our map for unit-only sensors, e.g. <code>min</code> → <code>mdi:timer-outline</code> .
7	domain	Our map, e.g. <code>light</code> → <code>mdi:lightbulb</code> .

Why `include_categories=primary` matters

HA's auto-generated boards list every entity a device exposes, most of which HA's own UI keeps behind the device page rather than on a dashboard. On the reference home the default board references 204 entities; 61 are **config** and 32 are **diagnostic**. The Deck view is nine entities, of which exactly one — `switch.deck_light` — is a thing a person would tap. The rest are the plug's LED toggle, auto-off timer, signal level and cloud-connection status.

Sending `include_categories=primary&include_hidden=false` takes that board from 204 tiles to 109. Note this is HA's own classification, not our opinion: some appliance integrations genuinely mark dozens of sensors as primary, and those areas stay busy.

Fallback tile rules (only when `entity_meta_available` is false)

Domain	Tile	Tap
light, switch, fan, cover, lock, climate, media_player, scene, script, automation, vacuum, button, input_boolean	Normal (bright)	Fires the device (\$5).
sensor, binary_sensor, number, select, camera, device_tracker, update, event, sun, zone, person, weather	Dimmed, read-only	Not tappable.

Errors — same as Screen 1, plus:

Status / code	When	App behavior
409 DASHBOARD_NOT_CONFIGURED	Board has no stored config — either it is still auto-generated, or the url_path does not exist . HA reports both the same way, so this is the code you get for a deleted board too.	Re-fetch /dashboards first. If the url_path is gone, treat it as deleted and go back to the list. If it is still listed, show “This board has no views yet — open it in Home Assistant, edit and save once.”
404 DASHBOARD_NOT_FOUND	Reserved. HA does not currently produce it — do not build the deleted-board path on this code.	—
400 VALIDATION	Unknown include_categories value.	Fix the request.

5. Action — Tap a tile

Tapping fires the entity's default action. Send the entity ID split into domain and suffix — `ha_entity` must NOT contain the dot.

Request

```
POST /automations/trigger
Authorization: Bearer <API_KEY>
Content-Type: application/json

{
  "home_id":      "scott_home",
  "ha_service":   "light",      // domain (before the dot)
  "ha_entity":    "desk_lamp",  // suffix only (after the dot)
  "user_ref":     "scott_mobile",
  "automation_id": "desk_lamp"  // same as ha_entity; enables the voice-gate check
}
```

Response — 200

```
{ "success": true, "message": "toggled light.desk_lamp",
  "status_code": 200, "latency_ms": 184 }
```

Response — 409, the voice gate

409 is not a failure. It means this entity is voice-gated for this user: the action is real, but it has to be authorized by speaking to the assistant. Mark the tile as gated and launch the VAPI session; never retry the trigger.

```
{ "error": "this automation requires voice authentication",
  "code": "ENROLLMENT_REQUIRED",
  "enrollment_id": "21e86f28-0037-40a9-94fe-cbdfa9fc6d07" }
```

The two 409 bodies are not the same. POST /automations/trigger returns only error, code and enrollment_id. POST /favorites/{id}/fire additionally returns automation_id, automation_name and home_id. The VAPI session needs automation_id and home_id — from the trigger path you already hold both (you just sent them), so use your own values rather than reading them off the response.

Never send a command to find out — ask

GET /voice-gated lists every entity that will answer 409 for a user. It is read-only, it fires nothing, and it runs the same lookup the fire path runs, so the two cannot disagree. Anything it lists will 409; anything it does not list will not.

```
GET /voice-gated?user_ref=scott_mobile&home_id=scott_home
Authorization: Bearer <API_KEY>

{
  "user_ref": "scott_mobile", "home_id": "scott_home", "count": 2,
  "items": [
    { "entity_id": "lock.yale_yrd226_tsdb", "automation_id": "yale_yrd226_tsdb",
      "name": "Yale Lock", "domain": "lock", "icon": "mdi:lock",
      "home_id": "scott_home", "enrollment_id": "9c520ce7-bdb5-...",
      "challenge_type": "STEP_UP", "created_by": "favorite_auto_lock" },
    { "entity_id": "scene.decorations_on", "automation_id": "decorations_on",
      "name": "Decorations On", "domain": "scene", "icon": "mdi:string-lights",
      "home_id": "scott_home", "enrollment_id": "21e86f28-0037-...",
      "challenge_type": "VERIFICATION", "created_by": "seed" }
  ]
}
```

Field	Meaning
entity_id	The entity that 409s. Compare against your tile's entity ID.
automation_id	The suffix the gate actually matches on. Any entity ending in .<automation_id> is gated, whatever its domain.
challenge_type	STEP_UP for locks, VERIFICATION otherwise. Informational.
enrollment_id	Pass to /enrollments/{id}/status to pause, or show in a “manage voice authentication” screen.
home_id	Which home the enrollment belongs to. Omit the query param to see all of the user's homes.

Only ACTIVE enrollments appear. A paused one does not gate and is not listed, so the list is always exactly the set that 409s right now.

Gating is per user, not per home. Two people in the same house get different answers, which is why user_ref is required here and matters on /dashboards/config. Always send the signed-in user's user_ref.

Where a gated entity can still appear, the row says so, so you never have to cross-reference by hand:

Surface	Gated entities	Field
GET /dashboards/config	Withheld by default — every tile is a button click	gated_excluded lists what was held back

Surface	Gated entities	Field
GET /items/search	Included	voice_gated on each row
GET /favorites	Included — this is where a lock lives	voice_auth_required on each row

Which entities return 409

A 409 comes from an **active enrollment** on (user_ref, automation_id). Enrollments are created two ways: automatically for every lock.* entity the moment it is favorited (locks can never be fired without the spoken challenge), and explicitly via POST /enrollments for anything else. Nothing else gates. If you have not enrolled it and it is not a lock, it will not 409.

Rule	Consequence for the app
The gate key is the entity suffix , not the full entity ID.	An enrollment on decorations_on gates both scene.decorations_on and script.decorations_on. Do not assume the domain narrows it.
The gate only runs when both user_ref and automation_id are sent.	Always send them. Omitting either fires the action ungated and skips the audit trail.
Locks auto-enroll on favorite.	Favoriting a lock.* entity always produces a gated <i>favorite</i> — and the lock stops being a board tile. The VAPI hand-off belongs on the Favorites screen, not the board.
Enrollments can be paused.	A PAUSED enrollment does not gate and is absent from /voice-gated, so the entity becomes a normal tile again. Do not cache “this is gated” across sessions.

Testing the 409 without touching the house

You should not need to. Read GET /voice-gated to know what is gated, and a board never hands you a gated tile in the first place. To exercise the 409 handler itself, the gate is evaluated before the command is dispatched and does not read home_id, so sending a home that does not exist proves the 409 path while making it impossible for anything to reach a real device. A gated automation_id returns 409; an ungated one falls through to dispatch and fails with 502, which is how you know the 409 came from the gate.

```
# 409 ENROLLMENT_REQUIRED — nothing is dispatched
curl -s -X POST "$BASE/automations/trigger" -H "Authorization: Bearer $KEY" \
-H "Content-Type: application/json" -d '{
  "home_id": "__no_such_home__", "ha_service": "scene", "ha_entity": "decorations_on",
  "user_ref": "scott_mobile", "automation_id": "decorations_on" }'

# control: an ungated automation_id reaches dispatch and 502s on the fake home
curl -s -X POST "$BASE/automations/trigger" -H "Authorization: Bearer $KEY" \
-H "Content-Type: application/json" -d '{
  "home_id": "__no_such_home__", "ha_service": "scene", "ha_entity": "good_night",
  "user_ref": "scott_mobile", "automation_id": "good_night" }'
```

To see the richer favorites 409 body, favorite the home's lock (POST /favorites with {"entity_id": "lock.<suffix>"}), then POST /favorites/{id}/fire. That also refuses before dispatch, so the lock never actuates.

Related read-only calls: GET /voice-gated?user_ref=<user> for the entity list (start here), GET /enrollments?user_ref=<user> for the raw enrollment rows including paused ones, and GET /check?user_ref=<user>&automation_id=<suffix> for one entity plus its cooldown and attempts-remaining.

Errors

Status / code	When	App behavior
400 VALIDATION	Missing field, or ha_entity contains a dot	Fix the request.
409 ENROLLMENT_REQUIRED	Entity is voice-gated for this user	Show “needs voice authentication”; launch the VAPI session.
502 (success: false)	HA rejected the command or the device is unreachable	Toast “Could not reach the device”; keep the tile enabled.

6. Caching and offline behavior

The server caches HA reads and serves stale data through brief outages, so the app does not need aggressive polling. Dashboard configs are cached 30 seconds, entity state 10 seconds, and the entity registry and icon translations 10 minutes — the last two only change when someone edits the home's HA. The app adds one layer:

- Cache the last good response of `/dashboards` and of each `/dashboards/config` (key: `url_path`), on disk.
- On `HOME_UNREACHABLE` (or any network error): render the cached copy, dim the grid, disable taps, and show a banner “Home unreachable — showing last known view.”
- On `entity_meta_available: false`: render the board, fall back to the domain icon table, and do *not* assume the category filter was applied.
- With no cache: show an empty state with a retry.
- Persist the last opened (board, view). On launch, open it directly and skip the boards list — the app opens on your own board.

7. Build order — step by step

Step	Build	Done when
1	HTTP client: base URL, Bearer header, error-envelope parsing into {error, code}	Any endpoint returns parsed JSON or a typed error.
2	Boards list screen from GET <code>/dashboards</code>	Rows match the home's HA sidebar; Overview first.
3	Board screen from GET <code>/dashboards/config</code> with <code>include_categories=primary&include_hidden=false</code> : tabs from <code>views[]</code> , tile grid from <code>entities</code>	Tabs show “title · count”; the Deck tab shows one tile, not nine.
4	Tiles rendered from <code>entity_meta</code> : icon, name, state; <code>controllable=false</code> renders read-only. Keep the domain table as the <code>entity_meta_available=false</code> fallback.	Signal-level and cloud-connection tiles are gone; the ones left look like HA.
5	Tile tap → POST <code>/automations/trigger</code> with split entity ID	A real light turns on; success toast shows.
6	Nothing to build on the board — gated entities are not tiles. Keep a 409 handler as the backstop, and use GET <code>/voice-gated</code> if a screen needs to name the voice-protected devices.	Every board tile fires. The 409 path is verified with the fake-home probe in §5, not by tapping real devices.
7	Disk cache + <code>HOME_UNREACHABLE</code> stale mode	Airplane-mode HA still shows last known screens.
8	Persist last (board, view); open on it at launch	Relaunch lands on the user's own room.

Live tile state now ships in `entity_meta[] . state`, so the tile model should carry it from the start. Still out of scope for this milestone: per-user board memory synced across devices.